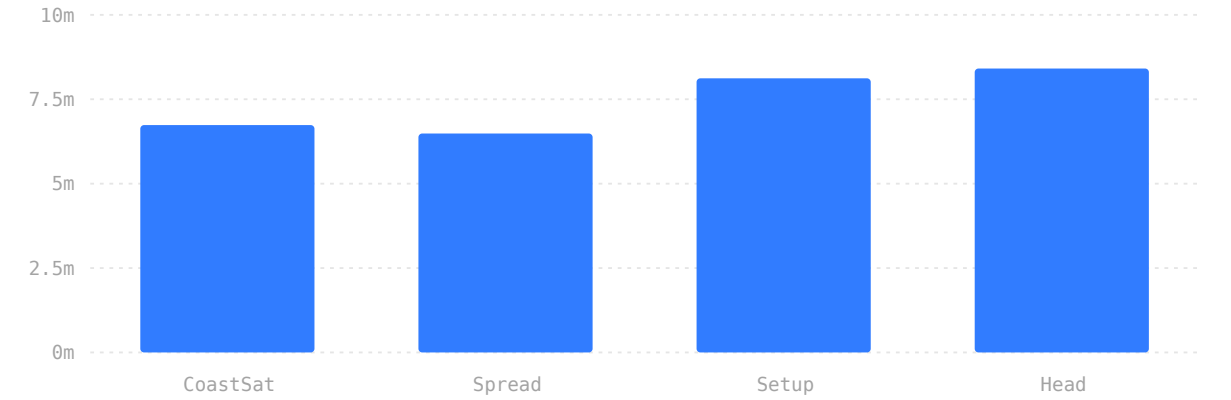
A coastal model worth training

The negative CPU screen, a concrete pretraining/post-training route, and the evidence needed before scaling.

Start with the result, not the architecture

> **The first screen did not support learning.** One beach, only 31 fitting scenes; newer sensors appeared later. The learned head made later measurements worse. This is not evidence that a bigger model is required.

Measured locally: Narrabeen, five transects; fit through 2009, calibration 2010–2016, test 2017–2019. Test: 329 transect rows from 68 acquisition groups. Retained: 48 scenes (70.6%); lower RMSE is better. No imagery, weights or external survey archives downloaded.



– Data table

Retained RMSE in metres, same test dates, separately ranked scenes

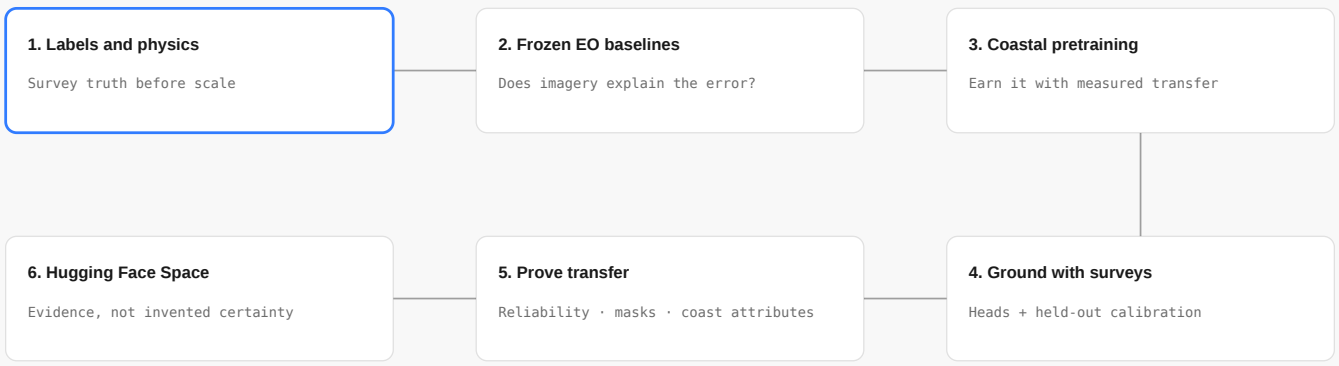
Route	RMSE
CoastSat	6.735 m
Spread	6.485 m
Setup	8.120 m
Head	8.410 m

CoastSat: published positions. Spread: simple disagreement ranking. Setup: calibration-chosen baseline. Head: learned metadata + disagreement. Primary comparison uses setup, not whichever baseline wins on test. Learned reduction: −3.57%; scene-bootstrap 95% interval −11.49% to +4.96%.

Coverage is not enough	95.1% marginal row coverage; 80.9% simultaneous scene coverage; mean interval width 36.7 m.
Useful change claims	0 of 138 exploratory pairs for metadata + disagreement. No claims is not a success.
Full Stage 0	Blocked: three sites' survey files absent; image QC/geometry inputs missing. No four-site or all-nine-baseline claim.

The path to a coastal measurement encoder

Everything below is proposed, not trained. Select a node for its decision gate.



1. Independent labels

Add bounded CC0 Truc Vert/Torrey surveys with approval. Audit datums, timestamps, rejected images and distinct scenes. Three sites are a screen, not foundation validation.

What pretraining should preserve

Proposed objectives and their failure modes

Learn	Do not erase or invent
Masked multispectral coastal structure	Boundaries and small coastal features; avoid mostly-empty-ocean training.
Tide/wave/sensor conditioning	Instantaneous waterline movement is real, even when a nuisance for fixed-datum change.
Geometry and missing-sensor handling	Registration on stable land, never on the shoreline; generated observations are not evidence.
Conditional temporal representation	Keep real erosion/recovery. “Same beach, all dates equal” destroys the target.

+ How we will judge “better”

The next decision is small

> **Proposed approval: up to 300 MB of external CC0 surveys.** Truc Vert monitoring surveys + grid: 162.61 MB. Torrey Pines mapped elevations: 115.22 MB. Total: 277.84 MB. Repeat the CPU screen across three sites; still no imagery, weights, GPU training or deployment. Expanded size and successful preprocessing not yet checked.

Exclude SWED and unresolved commercial imagery. Audit overlap: the 1,018-image Doodleverse dataset includes all four benchmark beaches. NatShore already has tide-aware selection and pixel confidence; coastal continual pretraining/adapters already exist. The potential contribution is independent-survey reliability and transferable physical-change evidence—not “first coastal foundation model.”

Sources: SDS_Benchmark · Truc Vert · Torrey surveys · Prithvi · TerraMind · NatShore · Coastal adapters.

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13 October 2026. Only the Narrabeen metrics were measured locally; all training stages and gates are proposals. Diagram is schematic. No model efficacy, new-site coverage or public release is established.